

Charts that answer questions

Follow-along worksheet

Your goal: choose a chart, read the result and explain one useful finding.
TRY IT means type along. WATCH means follow the demo first.

Name

A question I want a chart to answer

Before we start

Open follow-along.xlsx and go to mpg. Use this shorter route today.
Windows: select the cell > Formulas > Insert Python. Ctrl+Enter runs it.
Work in order. Later cells reuse mpg_df. Watching is welcome.

1 Count the cars in each group

TRY IT / mpg!K1. Type the code and run it.

```
mpg_df = xl("mpg[#All]", headers=True)
plt.figure()
sns.countplot(data=mpg_df, x="origin", color="#CF3338")
```

Which origin has the most cars?

How many?

Now add these lines to the end of K1 and rerun the whole cell:

```
plt.title("Cars in each group")
plt.xlabel("Origin")
plt.ylabel("Number of cars")
```

Check: USA has 207 cars. Image result? Right-click > Display Plot over Cells.

Three charts you can repeat

Work on mpg. Insert Python at each listed cell and run the whole block.

2 See the spread of fuel economy

TRY IT / K20. MPG means miles per gallon. Higher is better.

```
plt.figure()
sns.histplot(data=mpg_df, x="mpg", bins=10,
             color="#CF3338")
```

A histogram counts values within ranges. bins=10 makes ten ranges.

What does a taller histogram bar mean?

3 Compare weight with fuel economy

TRY IT / K40. Each dot is a car. Its color identifies origin.

```
plt.figure()
sns.scatterplot(data=mpg_df, x="weight", y="mpg",
               hue="origin",
               palette=["#CF3338", "#707070", "#1F2120"])
```

As weight increases, what tends to happen to MPG?

Explain the pattern to a colleague in one sentence

Result check

Heavier cars tend to have lower MPG in this sample.
The chart shows an association. Other factors can matter too.

Three more charts to explore

WATCH today. Keep this code for practice after the session.

4 Compare the middle and spread

mpg!K60. Box = middle half. Line in box = median. Whiskers usually extend to values within 1.5 box heights. Dots beyond may be unusual.

```
plt.figure()
sns.boxplot(data=mpg_df, x="origin", y="mpg",
            color="#CF3338")
```

Sample medians: USA 22.0 MPG, Europe 31.7, Japan 34.65.

5 Summarize numeric relationships

mpg!K80. Correlation runs from -1 to +1. A negative value means higher values of one measure tend to go with lower values of the other.

```
corr = mpg_df[["mpg", "weight", "cylinders"]].corr()
plt.figure()
sns.heatmap(corr, annot=True, fmt=".2f", cmap="Reds",
            vmin=-1, vmax=1)
```

Check: MPG and weight have correlation about -0.76. This does not prove cause.

6 Give each group its own chart

mpg!K100. col makes a separate chart for each origin, using the same axes. Model year 70 means 1970, and 82 means 1982.

```
sns.relplot(data=mpg_df, x="model_year", y="mpg",
            col="origin", color="#CF3338")
```

Which of these charts could help with your opening question?

Your next step

Use one small example to answer a question from your own work.

A question I want to answer in my own workbook

A familiar result or simple baseline I will check against

When I will practice

One question I still have

If you get stuck

Use this worksheet's cell locations and run steps in order.

#PYTHON!: check names, quotes and whether earlier cells ran.

#SPILL!: clear space for the result. #BUSY!: wait for calculation.

An Image result is normal. Right-click it > Display Plot over Cells.

Keep building with the book

Python in Excel for Data Analytics

George Mount / Packt

Chapters 3-4: visualization and exploration.

Practice with downloadable Excel workbooks
and adapt the examples to your questions.



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Help: Microsoft Python in Excel getting started guide